
Problem 9.12. Describe the error in the following fallacious "proof" that P $\neq$ NP. Assume that P = NP and obtain a contradiction. If P = NP, then $SAT \in$ P and so for some k, $SAT \in$ TIME(n^k). Because every language in NP is polynomial time reducible to SAT, you have NP $\subseteq$ TIME(n^k). Therefore, P $\subseteq$ TIME(n^k). But by the time hierarchy theorem, TIME(n^{k+1}) contains a language that isn't in TIME(n^k), which contradicts P $\subseteq$ TIME(n^k). Therefore, P $\neq$ NP.

Solution. The error is in argument "Because every language in NP is polynomial time reducible to SAT, you have NP $\subseteq$ TIME(n^k)". For NP $\subseteq$ TIME(n^k), every language in NP must be reducible to SAT in TIME(n^k) for some fixed k. □